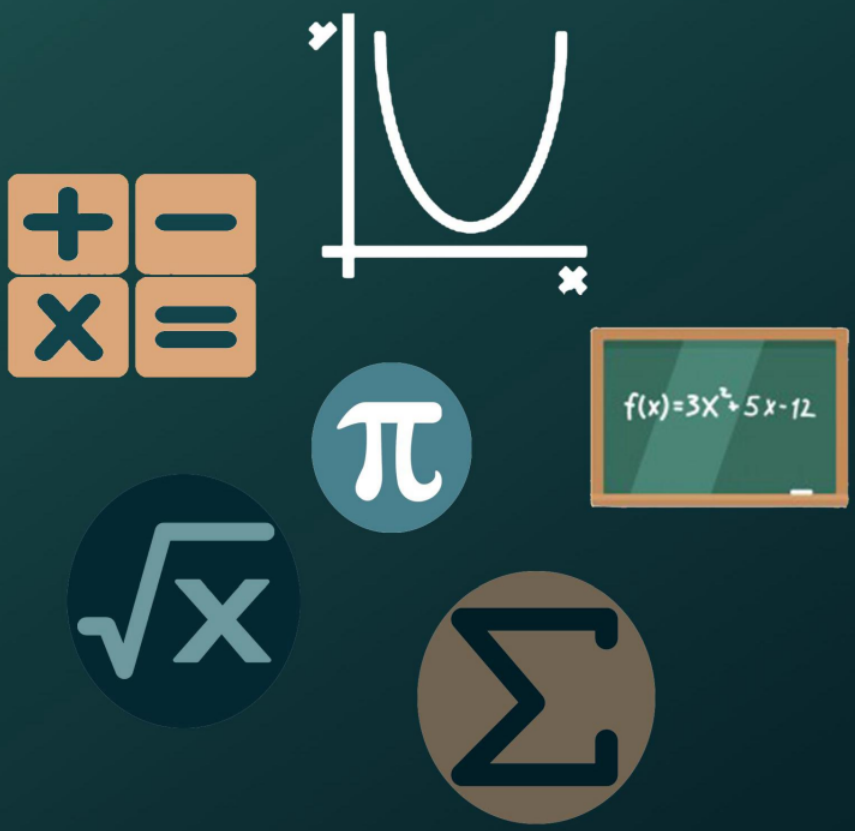
MATHEMATICS

Continuity



SHEET

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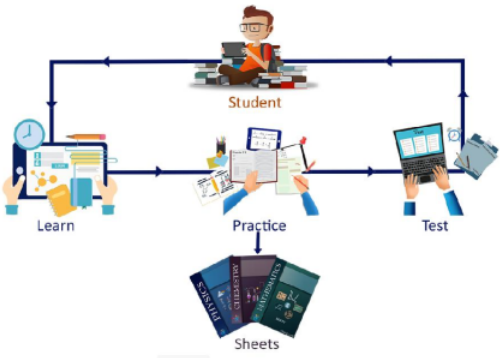
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Concept of Continuity				
2	Formulative Definition	1 -14	1, 2, 11 -17, 21 -25	1 - 6	1, 10, 11, 15 - 17
3	Reasons & Types of Discontinuities		3	8 - 11	6 - 8
4	Continuity in an Interval	15 -20	9, 18-19, 26	12 - 14	2, 18
5	Continuity of Functions defined by Functional Rule				
6	Continuity of Composite Functions		7 -8		3, 9, 12, 19
7	Intermediate Value Theorem (IMVT)				4, 13, 20, 21
8	Important Points about Continuity		4, 5		
9	Continuity over Countable Set	21	10, 20	15	5, 14, 22
10	After Complete Chapter		6		23 - 24

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Formulative Definition

Q1. If $f(x)=\begin{cases} x+\lambda, & x<3 \\ 4, & x=3 \\ 3x-5, & x>3 \end{cases}$ is continuous at $x = 3$, then λ is equal to :

(A) 3 (B) 2

(C) 1 (D) 4

Q2. If $f(x)=\dfrac{\log(1+ax)-\log(1-bx)}{x}$ is continuous at $x = 0$, then $f(0)$ is equal to :

(A) $\log a - \log b$ (B) $a + b$

(C) $a - b$ (D) ab

Q3. If $f(x)=\begin{cases} x\sin\dfrac{1}{x}, & x\neq 0 \\ k, & x=0 \end{cases}$ is continuous at $x = 0$, then the value of k is :

(A) 0 (B) 1

(C) –1 (D) 2

Q4. If $f(x)=\dfrac{2-\sqrt{x+4}}{\sin 2x}(x\neq 0)$ is continuous function at $x = 0$, then $f(0)$ equals :

(A) $\frac{1}{4}$ (B) $-\frac{1}{4}$

(C) $\frac{1}{8}$ (D) $-\frac{1}{8}$

Q5. If f(x) defined by $f(x)=\begin{cases} \dfrac{|x^2-x|}{x^2-x} & x\neq 0,1 \\ 1 & ,x=0 \\ -1 & ,x=1 \end{cases}$,

then f(x) is continuous for all -

(A) x

(B) x except at x = 0

(C) x except at x = 1

(D) x except at x = 0 and x = 1.

Q6. The points (s) of discontinuity of the function

$f(x)=\begin{cases} \dfrac{1}{5}(2x^2+3), & x\leq 1 \\ 6-5x, & 1<x<3 \\ x-3, & x\geq 3 \end{cases}$ is (are)

(A) x = 1 (B) x = 3

(C) x = 1, 3 (D) None of these

Q7. If $f(x)=\begin{cases} ax^2+b, & 0\leq x<1 \\ x+3, & 1<x\leq 2 \\ 4, & x=1 \end{cases}$, then the value of (a, b) for which f(x) cannot be continuous at x = 1 is

(A) (2, 2) (B) (3, 1)

(C) (4, 0) (D) (5, 2)

Q8. If function $f(x)=\dfrac{\sqrt{1+x}-\sqrt[3]{1+x}}{x}$, is continuous function, then f(0) is equal to -

(A) 2 (B) 1/4

(C) 1/6 (D) 1/3

Q9. If $f(x)=\begin{cases}\dfrac{x^2-(a+2)x+2a}{x-2} & , \quad x\neq 2\\ 2 & , \quad x=2\end{cases}$ is continuous at x = 2, then a is equal to -

(A) 0 (B) 1 (C) –1 (D) 2

Q10. If $f(x)=\begin{cases}[x]+[-x], & x\neq 2\\ \lambda, & x=2\end{cases}$, f is continuous at $x=2$ then λ is (where [.] denotes greatest integer)-

(A) –1 (B) 0 (C) 1 (D) 2

Q11. If $f(x)=\begin{cases}\dfrac{1-\cos 4x}{x^2} & , \quad x<0\\ a & , \quad x=0\\ \dfrac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4} & , \quad x>0\end{cases}$, then

correct statement is -

(A) f(x) is discontinuous at x = 0 for any value of a

(B) f(x) is continuous at x = 0 when a = 8

(C) f(x) is continuous at x = 0 when a = 0

(D) None of these

Q12. The function $f: R\sim\{0\}\to R$,

$f(x)=\dfrac{1}{x}-\dfrac{2}{e^{2x}-1}$ can be made continuous at x = 0 by defining f(0) as

(A) 2 (B) –1 (C) 0 (D) 1

Q13. If $f(x)=\begin{cases}x+2 & , \text{ when } x<1\\ 4x-1 & , \text{ when } 1\le x\le 3\\ x^2+5 & , \text{ when } x>3\end{cases}$,

then correct statement is -

(A) $\lim\limits_{x\to 1} f(x)=\lim\limits_{x\to 3} f(x)$

(B) f(x) is continuous at x = 3

(C) f(x) is continuous at x = 1

(D) f(x) is continuous at x = 1 and 3

Q14. If $f(x)=\begin{cases}\dfrac{1}{e^{1/x}+1} & , \quad x\neq 0\\ 0 & , \quad x=0\end{cases}$, then -

(A) $\lim\limits_{x\to 0+} f(x)=1$

(B) $\lim\limits_{x\to 0-} f(x)=0$

(C) f(x) is discontinuous at x = 0

(D) f(x) is continuous

Continuity in an Interval

Q15. If the function

$$f(x)=\begin{cases}1+\sin\dfrac{\pi}{2}x & \text{for } -\infty<x\le 1\\ ax+b & \text{for } 1<x<3\\ 6\tan\dfrac{x\pi}{12} & \text{for } 3\le x<6\end{cases}$$

is continuous in the interval (–∞, 6), then the values of *a* and *b* are respectivley :

(A) 1, 1 (B) 2, 0

(C) 0, 2 (D) 2, 1

Q16. For the function

$$f(x)=\frac{\log_e(1+x)+\log_e(1-x)}{x}$$

to be continuouss at $x=0$, the value of $f(0)$.

(A) –1 (B) 0

(C) –2 (D) 2

Q17. Let $f(x)=\begin{cases}\dfrac{\sqrt{1+px}-\sqrt{1-px}}{x}, & -1\le x<0\\ \dfrac{2x+1}{x-2}, & 0\le x\le 1\end{cases}$

If f(x) is continuous in the interval [–1, 1], then p equals :

(A) $\dfrac{1}{2}$ (B) $-\dfrac{1}{2}$

(C) –1 (D) 1

Q18. Let $f(x) = \frac{\sqrt{x^2+kx+1}}{x^2-k}$. The interval(s) of all possible values of k for which f is continuous for every $x \in R$, is -

(A) $(-\infty, -2]$ (B) $[-2, 0)$

(C) $R - (-2, 2)$ (D) $(-2, 2)$

Q19. Let $f(x) = \frac{1-\tan x}{4x-\pi}$, $x \neq \frac{\pi}{4}$, $x \in \left[0, \frac{\pi}{2}\right]$, if f(x) is continuous in $\left[0, \frac{\pi}{2}\right]$, then $f\left(\frac{\pi}{4}\right)$ is -

(A) –1 (B) $\frac{1}{2}$

(C) $-\frac{1}{2}$ (D) 1

Q20. Which of the following functions has finite number of points of discontinuity in R (where [.] denotes greatest integer).

(A) tan x

(B) $|x| / x$

(C) $x + [x]$

(D) $\sin [\pi x]$

Continuity over Countable Set

Q21. Function $f(x) = \frac{1}{\log |x|}$ is discontinuous at-

(A) One point

(B) Two points

(C) Three points

(D) Infinite number of points

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $f(x) = \lim_{n\to\infty} (\sin x)^{2n}$ then f is - ($n \in N$)

(A) continuous at $x = \frac{\pi}{2}$

(B) discontinuous at $x = \frac{\pi}{2}$

(C) discontinuous at $x = \frac{3\pi}{2}$

(D) discontinuous at infinite number of points

Q2. Which of the following function(s) not defined at x = 0 has/ have non- removable discontinuity at the origin-

(A) $f(x) = \frac{1}{1+2^{1/x}}$ (B) $f(x) = \tan^{-1}\frac{1}{x}$

(C) $f(x) = \frac{e^{1/x}-1}{e^{1/x}+1}$ (D) None of these

Q3. Which of the following function(s) not defined at x = 0 has/ have removable discontinuity at the origin-

(A) $f(x) = \frac{1}{1+2^{\cos x}}$

(B) $f(x) = \cos\left(\frac{|\sin x|}{x}\right)$

(C) $f(x) = x \sin\frac{\pi}{x}$

(D) $f(x) = \frac{1}{\ell n|x|}$

Q4. Which of the following statement (s) is(are) correct?

(A) If $f : R \to R$ be a continuous function which takes only rational values, then f is a constant function.

(B) If $f : [a, b] \to R$ be a continuous function then the range $\{f(x) : x \in [a, b]\}$ is a closed and bounded interval.

(C) The equation $x^{13} + 7x^3 - 5 = 0$ has a unique real root.

(D) The equation $(1 - x)\cos x = \sin x$ has no root in $(0, 1)$.

Q5. Which of the following statement(s) given below is/are incorrect?

(A) square of a discontinuous function is discontinuous function.

(B) $\lim\limits_{x \to 1} \log_x \sqrt{x}$ does not exist.

(C) The graph of the function $f(x) = \ln|x| - \ln x^2$ is symmetric about the y-axis.

(D) Inverse of a discontinuous function can be a continuous function.

Q.6 Which of the following function(s) defined below has/ have single point continuity.

(A) $f(x) = \begin{cases} 1 & \text{if } x \in Q \\ 0 & \text{if } x \notin Q \end{cases}$

(B) $g(x) = \begin{cases} x & \text{if } x \in Q \\ 1 - x & \text{if } x \notin Q \end{cases}$

(C) $h(x) = \begin{cases} x & \text{if } x \in Q \\ 0 & \text{if } x \notin Q \end{cases}$

(D) $k(x) = \begin{cases} x & \text{if } x \in Q \\ -x & \text{if } x \notin Q \end{cases}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 7 to 8)

Consider two real-valued functions $f(x)$ and $g(x)$ defined as follows.

$$f(x) = \begin{cases} x\left[\frac{1}{x}\right] + x[x], & x \neq 0 \\ 0 \quad , & x = 0 \end{cases}$$

and $g(x) = \sin x$, $\forall\, x \in R$.

where $[x]$ denotes greatest integer less than or equal to x.

Q.7 The function $h(x)$ defined by $h(x) = f(g(x))$ has

(A) $\lim\limits_{x \to \frac{5\pi}{4}} h(x) = \frac{3\sqrt{2}}{2}$.

(B) removable discontinuity of isolated type at $x = \frac{\pi}{6}$.

(C) irremovable discontinuity of finite type with jump 1 at $x = \frac{\pi}{6}$.

(D) non-removable discontinuity of finite type with jump $\frac{1}{2}$ at $x = \frac{\pi}{6}$.

Q.8 If $k(x) = \begin{cases} f(g(x)), & 0 < x < \frac{\pi}{2} \\ q \quad , & x = \frac{\pi}{2} \\ g(f(x)), & \frac{\pi}{2} < x < \pi \end{cases}$.

Then the value of q so that k(x) is continuous at $x = \frac{\pi}{2}$, is-

(A) 1

(B) $\frac{\sqrt{3}}{2}$

(C) $\frac{1}{2}$

(D) No such value of q exists.

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q.9 **Column-I** **Column-II**

(A) If $f(x) = \begin{cases} \sin\{x\}\,; & x<1 \\ \cos x + a\,; & x \geq 1 \end{cases}$ where {.} denotes the fractional part function, such that f(x) is continuous at x = 1. If $|k| = \dfrac{a}{\sqrt{2}\sin\dfrac{(4-\pi)}{4}}$ then k is (P) 1

(B) If the function $f(x) = \dfrac{(1-\cos(\sin x))}{x^2}$ is continuous at x = 0, then f(0) is (Q) 0

(C) $f(x) = \begin{bmatrix} x & , x \in Q \\ 1-x\,, & x \notin Q \end{bmatrix}$, then the values of x at which f(x) is continuous (R) –1

(D) If f(x) = x + {–x} + [x], where [x] and {x} represents integral and fractional part of x, then the values of x at which f(x) is discontinuous (S) $\frac{1}{2}$

Q10. **Column I** **Column II**

(A) If P(x) = [2 cos x], $x \in [-\pi,\pi]$, then P(x) (P) is discontinuous at exactly 7 points

(B) If Q(x) = [2 sin x], $x \in [-\pi, \pi]$ then Q(x) (Q) is discontinuous at exactly 4 points

(C) If R(x)= [2 tan x /2], $x \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, then R(x) (R) has non- removable discontinuities

(D) If $S(x) = \left[3\,\text{cosec}\frac{x}{3}\right]$, $x \in \left[\frac{\pi}{2},2\pi\right]$, then S(x) (S) is continuous at infinitely many values

SUBJECTIVE TYPE

Q11. $f(x)=\begin{cases}(\cos x)^{\cot^2 x} & x\neq 0\\ e^{-1/2} \quad if & x=0\end{cases}$ find whether the f (x) is continuous at x = 0 or not.

Q12. Find the value of a for which the following function is continuous at a given point.

$$f(x)=\begin{cases}\dfrac{x^2+3x-10}{x^2+2x-15}, & \text{when } x\neq -5\\ a, & \text{when } x=-5\end{cases}$$

at x = –5

Q13. Find the value of a for which the following function is continuous at a given point.

$$f(x)=\begin{cases}(\cos x)^{1/x}, & x\neq 0\\ a, & x=0\end{cases} \quad \text{at } x=0$$

Q14. Find the value of a for which the following function is continuous at a given point.

$$f(x)=\begin{cases}x+a & \text{if } x\le 1\\ 3-x^2 & \text{if } x>1\end{cases} \quad \text{at } x=1$$

Q15. Find the value of a for which the following function is continuous at a given point.

$$f(x)=\begin{cases}\dfrac{\sin[x]}{[x]+1}, & \text{for } x>0\\ \dfrac{\cos\frac{\pi}{2}[x]}{[x]}, & \text{for } x<0\\ a, & \text{at } x=0\end{cases} \quad \text{at } x=0,$$

where [x] denotes the greatest integer less than or equal to x.

Q16. Let $f(x)=\begin{cases}\dfrac{1-\sin\pi x}{1+\cos 2\pi x}, & x<\dfrac{1}{2}\\ p, & x=\dfrac{1}{2}\\ \dfrac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2}, & x>\dfrac{1}{2}\end{cases}$.

Determine the value of p, if possible, so that the function is continuous at x = 1/2.

Q17. Determine a & b so that f is continuous at $x=\frac{\pi}{2}$ where $f(x)=\begin{cases}\dfrac{1-\sin^3 x}{3\cos^2 x} & \text{if } x<\dfrac{\pi}{2}\\ a & \text{if } x=\dfrac{\pi}{2}\\ \dfrac{b(1-\sin x)}{(\pi-2x)^2} & \text{if } x>\dfrac{\pi}{2}\end{cases}$

Q18. Find the points of discontinuity of the following functions.

(i) $f(x)=\dfrac{1}{2\sin x-1}$

(ii) $f(x)=\dfrac{1}{x^2-3|x|+2}$

(iii) $f(x)=\dfrac{1}{x^4+x^2+1}$

Q19. Discuss the continuity of

$$f(x)=\lim_{n\to\infty}\frac{x^{2n}-1}{x^{2n}+1}$$

Q20. Discuss the continuity of f in [0,2] where $f(x)=\begin{cases}|4x-5|[x] & \text{for } x>1\\ [\cos\pi x] & \text{for } x\le 1\end{cases}$; where [x] is the greatest integer not greater than x. Also draw the graph.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q21. Find the value of (a – b) for which the following function is continuous at a given point

$$f(x)=\begin{cases}ax^2-b, & \text{when } 0\le x<1\\ 2, & \text{when } x=1\\ x+1, & \text{when } 1<x\le 2\end{cases} \quad \text{at } x=1$$

Q22. Find the value of (a – b) for which the following function is continuous at a given point.

$$f(x)=\begin{cases}\dfrac{x-4}{|x-4|}+a, & x<4\\ a+b, & x=4 \text{ at } x=4\\ \dfrac{x-4}{|x-4|}+b, & x>4\end{cases}$$

Q23. If the function $f(x)=\dfrac{3x^2+ax+a+3}{x^2+x-2}$ is continuous at $x=-2$. Find $f(-2)$.

Q24. Find the value of $(a+b)$ so that $f(x)$ is continuous for all $x \in R$ if

$$f(x)=\begin{cases}|ax+3| & \text{if } x\le -1\\ |3x+a| & \text{if } -1<x\le 0\\ \dfrac{b\sin 2x}{x}-2b & \text{if } 0<x<\pi\\ \cos^2 x-3 & \text{if } x\ge \pi\end{cases}$$

Q25. The function

$$f(x)=\begin{cases}\left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0<x<\frac{\pi}{2}\\ b+2 & \text{if } x=\frac{\pi}{2}\\ \left(1+|\cos x|\right)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2}<x<\pi\end{cases}$$

Determine the value of (a – 2b), if f is continuous at $x=\pi/2$.

Q26. $f(x)=\begin{cases}1+\sin\dfrac{\pi x}{2}, & \text{for } -\infty<x\le 1\\ ax+b, & \text{for } 1<x<3\\ 6\tan\dfrac{x\pi}{12}, & \text{for } 3\le x<6\end{cases}$

in the interval $(-\infty, 6)$. Find a & b if f(x) is continuous in $(-\infty, 6)$

Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The value of f(0), so that function,

$$f(x)=\frac{\sqrt{a^2-ax+x^2}-\sqrt{a^2+ax+x^2}}{\sqrt{a+x}-\sqrt{a-x}}$$

becomes continuous for all x, is given by - $(a\in R^+)$

(A) $a\sqrt{a}$ (B) $-\sqrt{a}$

(C) $\sqrt{a}$ (D) $-a\sqrt{a}$

Q2. If $f(x)=\dfrac{x-e^x+\cos 2x}{x^2}$, $x\ne 0$ is continuous at x = 0, then -

(A) $f(0)=\dfrac{5}{2}$ (B) $[f(0)]=-2$

(C) $\{f(0)\}=-0.5$ (D) $[f(0)].\{f(0)\}=-1.5$

where [x] and {x} denotes greatest integer and fractional part function.

Q3. Let $f(x)=\dfrac{x(1+a\cos x)-b\sin x}{x^3}$, $x\ne 0$ and $f(0)=1$. The value of a and b so that f is a continuous function are -

(A) 5/2, 3/2

(B) 5/2, –3/2

(C) –5/2, –3/2

(D) None of these

Q4. 'f' is a continuous function on the real line. Given that

$$x^2+(f(x)-2)x-\sqrt{3}.f(x)+2\sqrt{3}-3=0.$$

Then the value of $f(\sqrt{3})$ is -

(A) $\dfrac{2(\sqrt{3}-2)}{\sqrt{3}}$

(B) $2(1-\sqrt{3})$

(C) Zero

(D) Cannot be determined

Q5. The value of p for which the function

$$f(x)=\begin{cases}\dfrac{(4^x-1)^3}{\sin\dfrac{x}{p}\log\left[1+\dfrac{x^2}{3}\right]}, & x\neq 0\\ 12(\log 4)^3, & x=0\end{cases}$$

may be continuous at x = 0, is -

(A) 1 (B) 2

(C) 3 (D) 4

Q6. Let $g(x)=\tan^{-1}|x|-\cot^{-1}|x|$,

$f(x)=\dfrac{[x]}{[x+1]}\{x\}$, $h(x)=|g(f(x))|$

where {x} denotes fractional part and [x] denotes the integral part then which of the following holds good ?

(A) h is continuous at x = 0

(B) h is discontinuous at x = 0

(C) $h(0^-)=\pi/2$

(D) $h(0^+)=-\pi/2$

Q7. Consider $f(x)=\underset{n\to\infty}{\text{Limit}}\dfrac{x^n-\sin x^n}{x^n+\sin x^n}$ for x > 0, $x\neq 1$, f(1) = 0 then

(A) f is continuous at x = 1

(B) f has an infinite or oscillatory discontinuity at x = 1

(C) f has a finite discontinuity at x = 1

(D) f has a removable type of discontinuity at x = 1

Q8. Given $f(x)=\dfrac{\left[\{|x|\}\right]e^{x^2}\{[x+\{x\}]\}}{\left(e^{1/x^2}-1\right)\text{sgn}(\sin x)}$

for $x\neq 0 = 0$ for x = 0.

Where {x} is the fractional part function ; [x] is the step up function and sgn(x) is the signum function of x then, f(x) :-

(A) is continuous at x = 0

(B) is discontinuous at x = 0

(C) has a removable discontinuity at x = 0

(D) has an irremovable discontinuity at x = 0

Q9. Consider

$$f(x)=\begin{cases}x[x]^2\log_{(1+x)}2 & for\ -1<x<0\\ \dfrac{\ell n\left(e^{x^2}+2\sqrt{\{x\}}\right)}{\tan\sqrt{x}} & for\ 0<x<1\end{cases}$$

where [*] & {*} are the greatest integer function & fractional part function respectively, then :-

(A) $f(0)=\ell n2\Rightarrow$ f is continuous at x = 0

(B) $f(0)=2\Rightarrow$ f is continuous at x = 0

(C) $f(0)=e^2\Rightarrow$ f is continuous at x = 0

(D) f has an irremovable discontinuity at x = 0

Q10. Consider $f(x)=\dfrac{\sqrt{1+x}-\sqrt{1-x}}{\{x\}}$, $x\neq 0$;

$g(x)=\cos 2x$, $-\dfrac{\pi}{4}<x<0$,

$$h(x)-\begin{cases}\dfrac{1}{\sqrt{2}}f(g(x)) & \text{for } x<0\\ 1 & \text{for } x=0\\ f(x) & \text{for } x>0\end{cases}$$

then, which of the following holds good. where {x} denotes fractional part function.

(A) 'h' is continuous at x = 0

(B) 'h' is discontinuous at x = 0

(C) f(g(x)) is an even function

(D) f(x) is an even function

Q11. Consider the function defined on $[0,1]\to R$,

$f(x)=\dfrac{\sin x-x\cos x}{x^2}$ if $x\neq 0$ and f(0) = 0,

then the function f(x):-

(A) has a removable discontinuity at x = 0

(B) has a non removable finite discontinuity at x = 0

(C) has a non removable infinite discontinuity at x = 0

(D) is continuous at x = 0

Q12. Let $f : (1, \infty) \to (0, \infty)$ be a continuous decreasing function with $\lim_{x\to\infty} \frac{f(4x)}{f(8x)} = 1$. Then $\lim_{x\to\infty} \frac{f(6x)}{f(8x)}$ is equal to

(A) $\frac{4}{8}$ (B) $\frac{4}{6}$

(C) $\frac{6}{8}$ (D) 1

Q13. Given: $f(x) = \begin{cases} x & \text{for } |x| \le 1 \\ 1 & \text{for } |x| > 1 \end{cases}$,

$$g(x) = \begin{cases} \cos\left(\frac{\pi}{2}x\right) & \text{for } |x| \le 1 \\ |x-1| & \text{for } |x| > 1 \end{cases},$$

$$h(x) = \begin{cases} \frac{(|x|-1)}{\log_a |x|} & \text{for } |x| \ne 1 \\ \ln a & \text{for } |x| = 1 \; a > 0, a \ne 1 \end{cases}$$

If l, m, n denote the number of points of discontinuity of the functions f, g and h in their domain respectively, over R, then (l, m, n) is -

(A) (0, 0, 0) (B) (1, 1, 1)

(C) (2, 2, 2) (D) (1, 1, 0)

Q14. The function $f(x) = [x]^2 - [x^2]$ (where [y] is the greatest integer less than or equal to y), is discontinuous at :

(A) all integers

(B) all integers except 0 & 1

(C) all integers except 0

(D) all integers except 1

Q15. The function $f(x) = [x] . \cos \frac{2x-1}{2}\pi$, where $[\cdot]$ denotes the greatest integer function, is discontinuous at :-

(A) All x

(B) All integer points

(C) No x

(D) x which is not an integer

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Given $f(x) = \begin{cases} 3 - \left[\cot^{-1}\left(\frac{2x^3-3}{x^2}\right)\right] & \text{for } x > 0 \\ \{x^2\}\cos(e^{1/x}) & \text{for } x < 0 \end{cases}$

where { } & [ ] denotes the fractional part and the integral part functions respectively, then which of the following statement does not hold good-

(A) $f(0^-) = 0$

(B) $f(0^+) = 3$

(C) $f(0) = 0 \Rightarrow$ continuity of f at $x = 0$

(D) irremovable discontinuity of f at $x = 0$

Q2. If $f(x) = \sqrt{x}$ and $g(x) = x - 1$, then -

(A) fog is continuous on $[0, \infty)$

(B) gof is continuous on $[0,\infty)$

(C) fog is continuous on $[1, \infty)$

(D) None of these

Q3. Given the function $f(x) = \frac{1}{1-x}$, the points of discontinuity of the composite function $y = f(f(f(x)))$ are at $x =$

(A) 0 (B) 1 (C) 2 (D) –1

Q4. Which of the following statements is (are) correct for any function $g : [0, 2] \to [0, 2]$?

(A) If g is onto then g must be continuous.

(B) If g is continuous then g must be onto.

(C) g must be bounded.

(D) If g is continuous then g must be bounded.

Q5. Let $f(x) = \left[x\left[\frac{1}{x}\right]\right]$ for $x > 0$, where $[\cdot]$ denotes the greatest integer function, then f(x) is -

(A) Continuous at $x = 1/2$ if $f\left(\frac{1}{2}\right) = 0$

(B) Continuous at $x = 3/4$ if $f\left(\frac{3}{4}\right) = 0$

(C) discontinuous at finite number of points

(D) discontinuous at infinite number of points.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 6 to 8)

Consider a function, $f(x) = \frac{2}{3}\ell n^{3/2}(\sin x + \cos x)$ for $x \in \left(0, \frac{\pi}{2}\right)$.

Q6. The value of $\lim\limits_{x\to\frac{\pi}{2}} \frac{f(x)}{\left(\frac{\pi}{2}-x\right)^{3/2}}$ is :-

(A) is $\frac{2}{3}$ (B) is 1

(C) is $\frac{2}{3}(2)^{3/4}$ (D) Non existent

Q7. The function g(x) is defined as

$$g(x) = \begin{bmatrix} \left(\frac{3}{2}f(x)\right)^{2/3} & x \in \left(0, \frac{\pi}{2}\right) \\ \frac{e^{\left(\frac{3}{2}f\left(x-\frac{\pi}{2}\right)\right)^{2/3}}}{\sqrt{2}}; & x \in \left(\frac{\pi}{2}, \pi\right) \end{bmatrix}.$$

Then g(x) -

(A) is continuous at $x = \frac{\pi}{2}$ for $g\left(\frac{\pi}{2}\right) = 0$

(B) has a removable discontinuity at $x = \frac{\pi}{2}$

(C) g(x) is discontinuous at $x = \frac{\pi}{2}$ and jump of discontinuity is equals to $\sqrt{2}$

(D) has a non-removable discontinuity at $x = \frac{\pi}{2}$

Q8. The range of g(x) is (g(x) same as **Q.7**) :-

(A) (0, 1] (B) $\left(0, \frac{1}{2}\ell n2\right]$

(C) $\left(\frac{1}{2}, 1\right]$ (D) $\left(0, \frac{1}{2}\ell n2\right] \cup \left(\frac{1}{\sqrt{2}}, 1\right]$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9.

Column-I	**Column-II**
(A) $f(u) = \frac{1}{u^2+u-2}$, where $u = \frac{1}{x-1}$. The values of x at which 'f' is discontinuous.	(P) $\frac{1}{2}$
(B) $f(x) = u^2$, where $u = \begin{cases} x-1, x \geq 0 \\ x+1, x < 0 \end{cases}$ The number of values of x at which 'f' is discontinuous.	(Q) 0
(C) If the function $f(x) = \frac{(1-\cos(\sin x))}{x^2}$ is continuous at $x = 0$, then f(0) is -	(R) 2
(D) $f(x) = \begin{bmatrix} x & , x \in Q \\ 1-x & , x \notin Q \end{bmatrix}$, then the values of x at which f(x) is continuous.	(S) 1

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q10. If $\sum_{r=1}^{n} \sin^{-1}\alpha_r = \frac{n\pi}{2}$ for any $n \in N$

and $p = \prod_{r=1}^{n} (\alpha_r)^r$.

If $f(x) = \begin{cases} \frac{x^{1/3} - (3-2x)^{1/4}}{x^2 - x}, & x \neq p \\ k, & x = p \end{cases}$ is

continuous at x = p, then find the value of 6(k + p).

Q11. If $f(x) = \begin{cases} \frac{\sin 2x - \tan 2x}{(e^x - 1)x^2} & \text{if } x \neq 0 \\ K & \text{if } x = 0 \end{cases}$ is

continuous at x = 0 then find the absolute value of K.

Q12. If the function

$$f(x) = \frac{\tan(\tan x) - \sin(\sin x)}{\tan x - \sin x} \quad (x \neq 0)$$

is continuous at x = 0, then find the value of f (0).

Q13. Let f be a real valued continuous function on R and satisfying $f(-x) - f(x) = 0\ \forall\ x \in R$. If f (– 5) = 5, f (– 2) = 4, f (3) = – 2 and f(0) = 0 then find the minimum number of zero's of the equation f(x) = 0.

Q14. Find the number of points of discontinuity of the function f(x) = [5x] + {3x} in [0, 5]. Where [y] and {y} denote largest integer less than or equal to y and fractional part of y respectively.

SUBJECTIVE TYPE

Q15. If $f(x) = \begin{cases} \frac{(e^x - 1)^3 \operatorname{cosec}(ax)}{\ln(1+x^2)} & x \neq 0 \\ b & x = 0 \end{cases}$

is continuous, find b.

Q16. Find the value of f (0) for which the following function is continuous at x = 0.

$$f(x) = \frac{\sqrt{a^2 - ax + x^2} - \sqrt{a^2 + ax + x^2}}{\sqrt{a+x} - \sqrt{a-x}} \quad (a > 0)$$

Q17. Consider the function

$$g(x) = \begin{cases} \frac{1 - a^x + xa^x \ell na}{a^x x^2} & \text{for } x < 0 \\ \frac{2^x a^x - x\ell n2 - x\ell na - 1}{x^2} & \text{for } x > 0 \end{cases}$$

where a > 0.

Find the value of 'a' & 'g(0)' so that the function g(x) is continuous at x = 0.

Q18. Find the set of values of x, where f(x) is continuous

$$f(x) = \lim_{n\to\infty} \frac{\log(2+x) - x^{2n}\sin x}{1 + x^{2n}}$$

Q19. Let $g(x) = \lim_{n\to\infty} \frac{x^n f(x) + h(x) + 1}{2x^n + 3x + 3}$, $x \neq 1$ and

$g(1) = \lim_{x\to 1} \frac{\sin^2(\pi \cdot 2^x)}{\ell n\left(\sec(\pi \cdot 2^x)\right)}$ be a continuous

function at x = 1, find the value of

4 g (1) + 2f(1) – h(1). Assume that f(x) and h(x) are continuous at x = 1.

Q20. Suppose that f (x) is continuous in [0, 1] and f (0) = 0, f (1) = 0.

Prove that $f(c) = 1 - 2c^2$ for some $c \in (0, 1)$.

Q21. Let f be continuous on the inteval [0, 1] to R such that f (0) = f (1).

Prove that there exists a point c in $\left[0, \frac{1}{2}\right]$ such that $f(c) = f\left(c + \frac{1}{2}\right)$.

Q22. Find the points of discontinuity of the following functions.

(i) $f(x) = \frac{1}{1 - e^{\frac{x-1}{x-2}}}$

(ii) f(x) = [[x]] – [x – 1], where [·] represents the greatest integer function.

Q23. Let $f(x) = \underset{n\to\infty}{\text{Lim}} \frac{x^{2n-1} + ax^3 + bx^2}{x^{2n} + 1}$. If f(x) is continuous for all x ∈ R, find the bisector of angle between the lines 2x + y – 6 = 0 and 2x – 4y + 7 = 0 which contains the point (a,b).

Q24. Given, $f(x) = \sum_{r=1}^{n} \tan\left(\frac{x}{2^r}\right) \sec\left(\frac{x}{2^{r-1}}\right)$; r, n ∈ N

g(x) =

$$\underset{n\to\infty}{\text{Limit}} \frac{\ell n\left(f(x) + \tan\frac{x}{2^n}\right) - \left(f(x) + \tan\frac{x}{2^n}\right)^n \cdot \left[\sin\left(\tan\frac{x}{2}\right)\right]}{1 + \left(f(x) + \tan\frac{x}{2^n}\right)^n}$$

= k for $x = \frac{\pi}{4}$ and the domain of g(x) is (0, π/2).

where [ ] denotes the greatest integer function.

Find the value of k, if possible, so that g(x) is continuous at x = π/4. Also state the points of discontinuity of g(x) in (0, π/4), if any.

Answer Key

Ex -1

1	2	3	4	5	6	7	8	9	10
C	B	A	D	D	B	D	C	A	A
11	12	13	14	15	16	17	18	19	20
B	D	C	C	B	B	B	B	C	B
21									
C									

Ex - 1A

1. B,C,D **2.** A,B,C **3.** B,C,D **4.** ABC **5.** A,B,D

6. B,C, D **7.** D **8.** A

9. (A) → (P,R); (B) → (S); (C) → (S); (D) → (P, Q, R)

10. (A) → (P,R,S); (B) → (P,R,S); (C) → (Q,R,S); (D) → (R,S)

11. f(x) is continuous at x = 0. **12.** a = 7/8 **13.** a = 1 **14.** a = 1

15. a = 0 **16.** No value of P **17.** a = 1/2, b = 4

18. (i) $x = 2n\pi + \frac{\pi}{6}$ or $x = 2n\pi + \frac{5\pi}{6}$, n ∈ Z **(ii)** x = ±1, ±2 **(iii)** x ∈ R

19. (iii) Novalue x, where f(x) is discontinuous

20. the function f is continuous everywhere in [0, 2] except for $x = 0, \frac{1}{2}, 1$ & 2.

21. 2 **22.** 2 **23.** -1 **24.** 1 **25.** 2

26. a = 2, b = 0

Ex - 2									
1	2	3	4	5	6	7	8	9	10
B	D	C	B	D	A	C	A	D	A
11	12	13	14	15					
D	D	D	D	C					

Ex - 2A

1. [B,D] **2.** [B,C] **3.** [A,B] **4.** CD **5.** [B,D]

6. A **7.** D **8.** D

9. (A) $\rightarrow$ (P,R,S); (B) $\rightarrow$ (Q); (C) $\rightarrow$ (P); (D) $\rightarrow$ (P)

10. 0011 **11.** 0004 **12.** 0002 **13.** 5 **14.** 30

15. $b = \frac{1}{a}$ **16.** $-\sqrt{a}$ **17.** $a = \frac{1}{\sqrt{2}}$, $g(0) = \frac{(\ell n 2)^2}{8}$

18. $R - \{-1, 1\}$ **19.** 5

20. Answer not required **21.** Answer not required

22. (iv) $x = 2$ and $x = 1$

(v) $f(x)$ is continuous $\forall\ x \in R$.

23. $6x - 2y - 5 = 0$ **24.** $k = 0$; $g(x)$ is continuous everywhere.

-: Important Notes & Formulas :-